디지털 논리

Application of Boolean Algebra

이상화

Review: How to Manipulate (1)

1. Combining terms

$$\boxed{XY + XY' = X}$$

$$abc'd' + abcd' = abd' \qquad [X = abd', Y = c]$$

2. Adding terms using

$$\boxed{X + X = X}$$

$$ab'c + abc + a'bc = ab'c + abc + \boxed{abc} + a'bc = ac + bc$$

Review: How to Manipulate (2)

3. Eliminating terms

$$\boxed{X + XY = X}$$

$$a'b + a'bc = a'b \qquad [X = a'b]$$

$$\boxed{XY + X'Z + YZ = XY + X'Z}$$

$$a'bc' + bcd + a'bd = a'bc' + bcd$$

$$[X = c, Y = bd, Z = a'b]$$

Review: How to Manipulate (3)

4. Eliminating literals

$$\boxed{X + X'Y = X + Y}$$

$$\begin{aligned}
A'B + A'B'C'D' + ABCD' &= A'(B + B'C'D') + ABCD' \\
&= A'(B + C'D') + ABCD' \\
&= B(A' + ACD') + A'C'D' \\
&= B(A' + CD') + A'C'D' \\
&= A'B + BCD' + A'C'D'
\end{aligned}$$

Review: How to Manipulate (4)

5. Adding redundant terms
- adding *xx'*, multiplying (*x*+*x'*)
- adding *yz* to *xy* + *x'z*, adding *xy* to *x*, …

$$\underline{WX} + XY + \underline{X'Z'} + WY'Z'$$

$$= WX + XY + X'Z' + WY'Z' + WZ'$$ (add *WZ'* by consensus theorem)

$$= \underline{WX} + XY + \underline{X'Z'} + \underline{WZ'}$$ (eliminate *WY'Z'*)

$$= WX + XY + X'Z'$$ (eliminate *WZ'*)

Ex 1: Design of Digital Circuit

Given word description

1. The alarm will ring iff

- the alarm switch is turned on ***and*** the door is not closed,

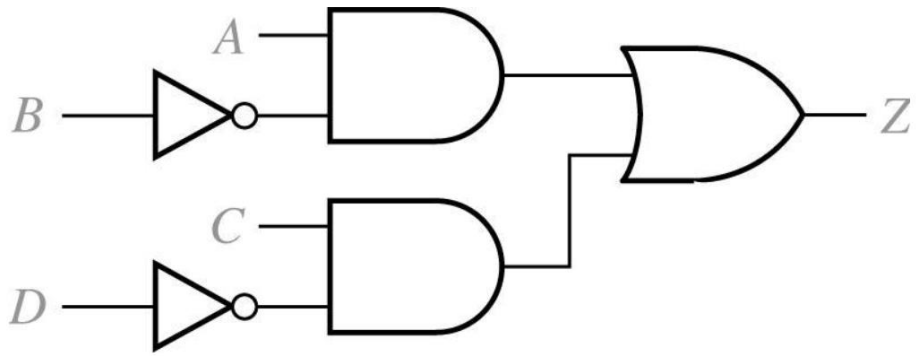
- ***or*** it is after 6PM ***and*** window is not closed.

Logic assignment => ring: Z, alarm switch: A, door: B, after 6PM : C, window: D

2. Boolean expression

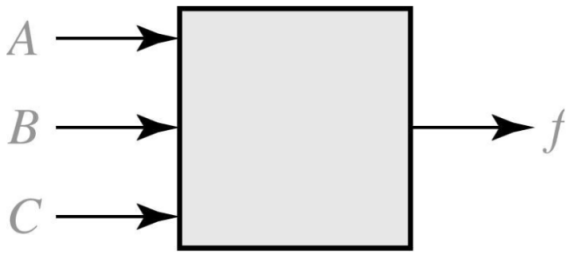
$$Z = AB' + CD'$$

3. Circuit realization



Ex 2: Design of Digital Circuit (1)

Given Truth Table



A B C	f	f'
0 0 0	0	1
0 0 1	0	1
0 1 0	0	1
0 1 1	1	0
1 0 0	1	0
1 0 1	1	0
1 1 0	1	0
1 1 1	1	0

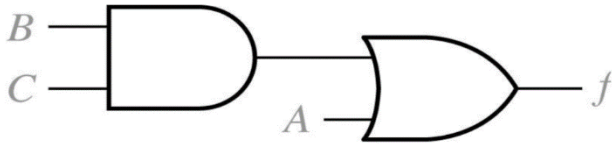
Logic (Boolean) expression for *f=1*

$$f = A'BC + AB'C' + AB'C + ABC' + ABC$$

Simplified equation of sum of product form

$$f = A'BC + AB' + AB = A'BC + A = A + BC$$

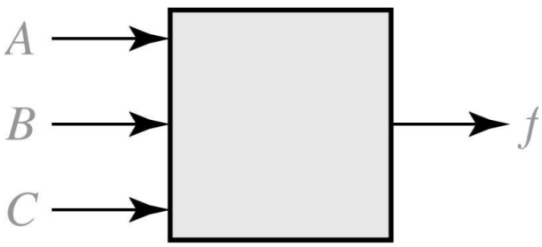
Circuit realization



$$\boxed{X + X'Y = X + Y}$$

Ex 2: Design of Digital Circuit (2)

Truth Table



A	B	C	f	f'
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

Logic expression for $f = 0$ ($f' = 1$) and take the complement of f'

$$f' = A'B'C' + A'B'C + A'BC'$$

$$\rightarrow \quad f = (A+B+C)(A+B+C')(A+B'+C)$$

Simplified equation of product of sum form

$$f = (A+B)(A+B'+C) = A + B(B'+C)$$
$$= A + BC = (A+B)(A+C)$$

Minterm

- **Product term: XY, WZ'**
- **Sum-of-product expression: X'Y + XY'Z'**
- **Minterm (m)**
  - Every variable appears once in a product
  - WXYZ, WXYZ', …
  - Each minterm has the output of 1.
  - Minterm expansion
    - Standard sum of products
    - For every output = 1 in the truth table, sum of every minterm

$$f = A'BC + AB'C' + AB'C + ABC' + ABC$$

Maxterm

- **Sum terms: X+Y, X'+Y+Z'**
- **Product-of-sum expression: (X+Y)(W+Z)(X'+Y'+Z)**
- **Maxterm (M)**
  - Every variable appears once in a sum
  - W+X+Y+Z, …
  - Each maxterm has the output of 0
  - Maxterm expansion
    - Standard product of sums
    - For every output= 0 in the truth table, product of every maxterm

$$f = (A+B+C)(A+B+C')(A+B'+C)$$

- **Maxterm is the complement of minterm.**
  - Given a minterm or maxterm expansion, we know the other expansion.

Minterms vs Maxterms (I)

Row	X	Y	Z	F		Minterm	Maxterm
0	0	0	0	F(0,0,0)		$X' \cdot Y' \cdot Z'$	$X + Y + Z$
1	0	0	1	F(0,0,1)	0	$X' \cdot Y' \cdot Z$	$X + Y + Z'$
2	0	1	0	F(0,1,0)		$X' \cdot Y \cdot Z'$	$X + Y' + Z$
3	0	1	1	F(0,1,1)		$X' \cdot Y \cdot Z$	$X + Y' + Z'$
4	1	0	0	F(1,0,0)		$X \cdot Y' \cdot Z'$	$X' + Y + Z$
5	1	0	1	F(1,0,1)	1	$X \cdot Y' \cdot Z$	$X' + Y + Z'$
6	1	1	0	F(1,1,0)		$X \cdot Y \cdot Z'$	$X' + Y' + Z$
7	1	1	1	F(1,1,1)		$X \cdot Y \cdot Z$	$X' + Y' + Z'$

$$f(A,B,C) = m_3 + m_4 + m_5 + m_6 + m_7$$
$$= \sum m(3,4,5,6,7)$$

Minterm expansion

$$f(A,B,C) = M_0 M_1 M_2$$
$$= \prod M(0,1,2)$$

Maxterm expansion

Minterms vs Maxterms (II)

- Maxterm is the complement of minterm.
  - Given a minterm or maxterm expansion, we know the other expansion.

$$F = (X'Y'Z' + X'YZ' + XY'Z + XYZ)$$
$$= m_0 + m_2 + m_5 + m_7$$

$$F(X,Y,Z) = \sum m(0,2,5,7)$$

$$F(X,Y,Z) = \prod M(1,3,4,6)$$

Properties of Minterm

1. **2^n minterms for *n* Boolean variables.**
2. **Any Boolean function can be expressed as a logical sum of minterms.**
3. **The complement of a function contains those minterms not included in the original function.**
4. **A function that includes all the 2^n minterms is equivalent to logic 1.**
5. **Orthogonality of minterms**
   - each minterm has at least the complement of one of variables.

$$m_i m_j = 0, \ i \neq j$$

- Maxterms are seldom used!
  =>To be multiplied out!

Finding Minterms from Simplification

Multiplying all missing variables (X+X')(Y+Y') for each product term

$$f(a,b,c,d)$$

$$= a'(b'+d)+acd'$$

$$= a'b'+a'd+acd'$$

$$= a'b'(c+c')(d+d')+a'd(b+b')(c+c')+acd'(b+b')$$

$$= \underset{m_0}{a'b'c'd'}+\underset{m_1}{a'b'c'd}+\underset{m_2}{a'b'cd'}+\underset{m_3}{a'b'cd}+\underset{m_5}{a'bc'd}+\underset{m_7}{a'bcd}+\underset{m_{14}}{abcd'}+\underset{m_{10}}{ab'cd'}$$

$$= \sum m(0,1,2,3,5,7,10,14)$$

$$= \prod M(4,6,8,9,11,12,13,15)$$

Finding Maxterms

Adding all missing variables (XX'+YY'….) in each sum

$$f(a,b,c,d)$$
$$= a'(b'+d)+acd'$$
$$= (a'+cd')(a+b'+d)$$
$$= (a'+c)(a'+d')(a+b'+d)$$
$$= (a'+bb'+c+dd')(a'+bb'+cc'+d')(a+b'+cc'+d)$$
$$= (a'+bb'+c+d)(a'+bb'+c+d')(a'+bb'+c+d')(a'+bb'+c'+d')(a+b'+cc'+d)$$
$$= (a'+b+c+d)(a'+b'+c+d)(a'+b+c+d')(a'+b'+c+d')$$
$$(a'+b+c'+d')(a'+b'+c'+d')(a+b'+c+d)(a+b'+c'+d)$$
$$= \prod M(4,6,8,9,11,12,13,15)$$
$$= \sum m(0,1,2,3,5,7,10,14)$$

General Minterm and Maxterm Expansion

For all values in truth table of n variables

$$F = \sum_{i=0}^{2^n-1} a_i m_i = \prod_{i=0}^{2^n-1} (a_i + M_i)$$

$$a_i = 0 \text{ or } 1$$

By orthogonality of minterms

$$F_1 = \sum_{i=0}^{2^n-1} a_i m_i, \quad F_2 = \sum_{i=0}^{2^n-1} b_i m_i$$

$$F_1 F_2 = \sum_{i=0}^{2^n-1} a_i b_i m_i$$

Incomplete Functions

When we have the don't care (X) for some circuits, the don't cares are optional for minterm/maxterm expansions

$$F = A'B'C' + A'BC + ABC = A'B'C' + BC$$

$$F = A'B'C' + \boxed{A'B'C} + A'BC + ABC = A'B' + BC$$

$$F = A'B'C' + \boxed{A'B'C} + A'BC + \boxed{ABC'} + ABC = A'B' + BC + AB$$

A B C	F
0 0 0	1
0 0 1	X
0 1 0	0
0 1 1	1
1 0 0	0
1 0 1	0
1 1 0	X
1 1 1	1

General expression for incomplete functions

$$F = \sum m(0,3,7) + \sum d(1,6)$$

$$= \prod M(2,4,5) \prod D(1,6)$$

Ex: Design of Binary Adder

Ex1: Binary adder with two 1-bit inputs and 2-bit output

- A, B : 1 bit inputs (A+B)
- X: carry output
- Y: adding output

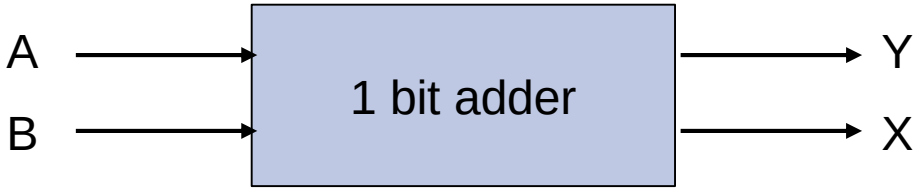
SUM

$$\begin{array}{r} A \\ +\ B \\ \hline X\ Y \end{array}$$

A B	sum	X Y
0 0	0 0	0 0
0 1	0 1	0 1
1 0	0 1	0 1
1 1	1 0	1 0

$$X = AB$$

$$Y = A'B + AB' = A \oplus B$$



Ex: Design of Error Detector

Error detector for 6-3-1-1 BCD digits.

=> The output(F) is to be 1 iff the four inputs (A,B,C,D) represent an invalid code combination.

A B C D	F
0 0 0 0	0
0 0 0 1	0
0 0 1 0	1
0 0 1 1	0
0 1 0 0	0
0 1 0 1	0
0 1 1 0	1
0 1 1 1	0
1 0 0 0	0
1 0 0 1	0
1 0 1 0	1
1 0 1 1	0
1 1 0 0	0
1 1 0 1	1
1 1 1 0	1
1 1 1 1	1

중복되는 것 중 큰 수는 error

10 이상 큰 수는 error (0~9 표현)

$$F = \sum m(2,6,10,13,14,15)$$
$$= A'B'CD' + A'BCD' + AB'CD' + ABCD' + ABC'D + ABCD$$
$$= A'CD' + ACD' + ABD$$
$$= CD' + ABD$$

- 3 gates circuit

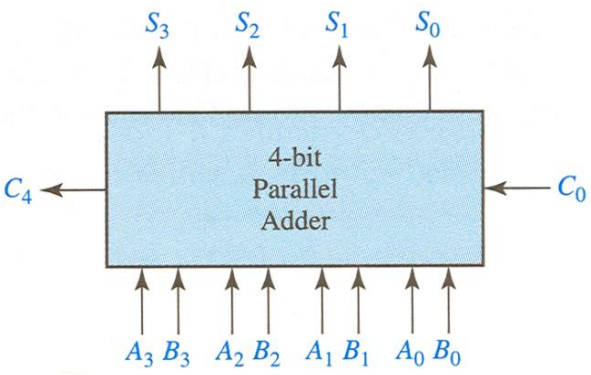
=> 2-input AND gate, 3-input AND gate, and 2-input OR gate

Design of Parallel Binary Adder (I)

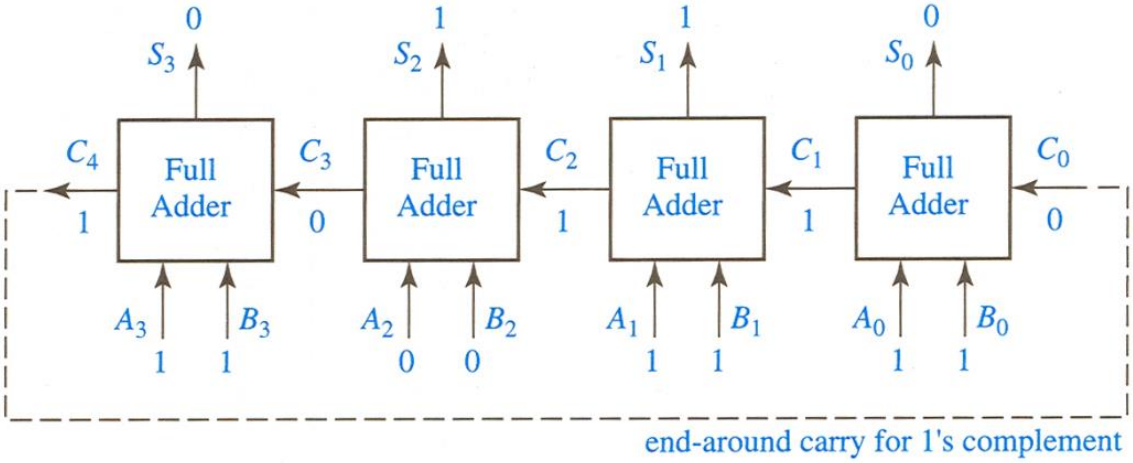
Design a **parallel** adder of **4-bit unsigned** binary numbers and a carry input.

- **9 input** variables (two 4-bit inputs and a carry input)
- **5 output** variables (4-bit output and a carry output)

=> Design a **full adder** to add two 1-bit inputs and **a carry input**, and then connect this modules to form 4-bit adder



4-bits parallel adder



4-bits parallel adder with 4 full adders

Design of Parallel Binary Adder (II)

Design the full adder using truth table

- Inputs: two 1 bit inputs and a input carry bit
- Outputs: 1 bit sum, 1 carryout bit

$$sum = X'Y'C_{in} + X'YC_{in}' + XY'C_{in}' + XYC_{in}$$
$$= X'(Y'C_{in} + YC_{in}') + X(Y'C_{in}' + YC_{in})$$
$$= X'(Y \oplus C_{in}) + X(Y \oplus C_{in})'$$
$$= X \oplus Y \oplus C_{in}$$

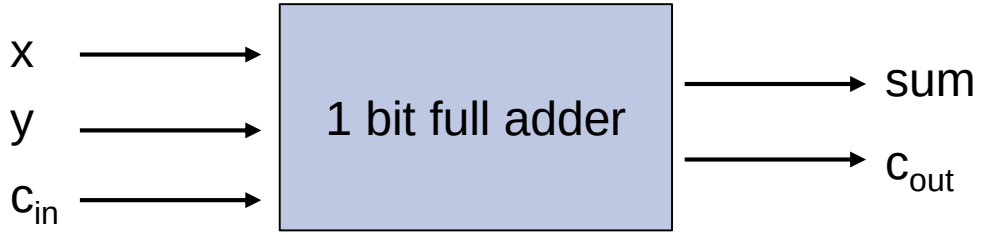
X	Y	C_{in}	C_{out}	sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

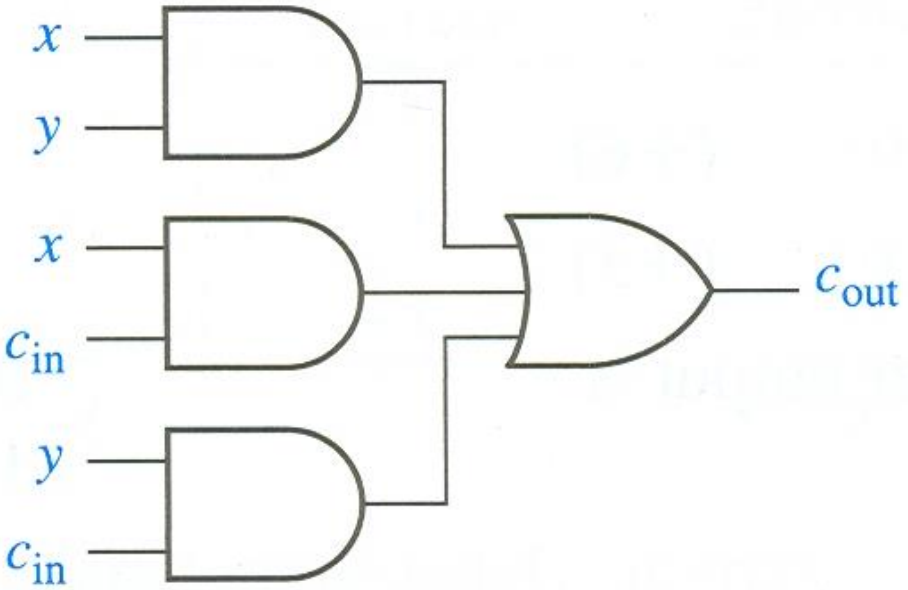
Full adder truth table

$$C_{out} = X'YC_{in} + XY'C_{in} + XYC_{in}' + XYC_{in}$$
$$= (X'YC_{in} + XYC_{in}) + (XY'C_{in} + XYC_{in}) + (XYC_{in}' + XYC_{in})$$
$$= YC_{in} + XC_{in} + XY$$

Design of Parallel Binary Adder (III)

Design the full adder using gates





Design of Parallel Binary Adder (IV)

Design the adder of **negative numbers** (4bits inputs: $A_3A_2A_1A_0 + B_3B_2B_1B_0$)

=> same operation in **2's complement** binary system

=> there is no carry input of the first full adder

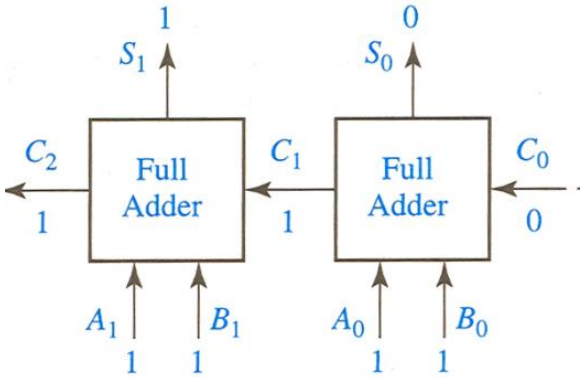
=> discard the last carry output

=> overflow signal (V) - A_3, B_3, S_3 are sing bits

$$V = A_3^{'}B_3^{'}S_3 + A_3B_3S_3^{'}$$

Positive sum
=> negative

Negative sum
=> positive

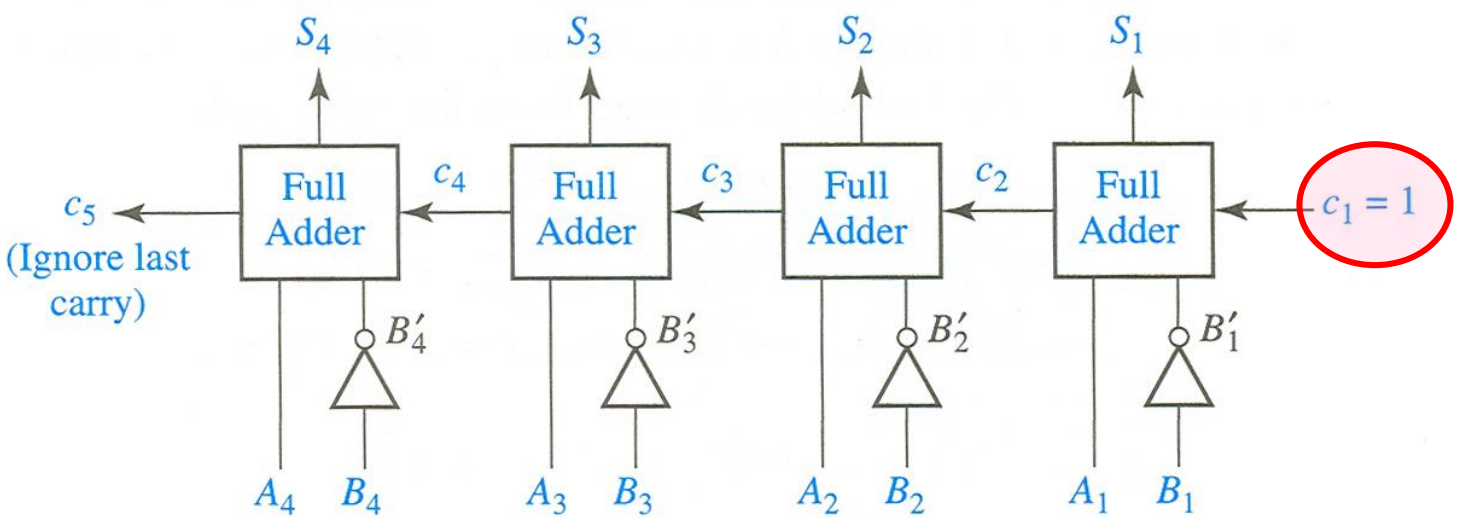


Design of Binary Subtracter (I)

1) Design the subtracter using full adder (4 bits inputs: $A_3A_2A_1A_0$ - $B_3B_2B_1B_0$)

=> add with negative number in **2's complement**

=> first 1's complement (inverter) and then add 1 (C_1)



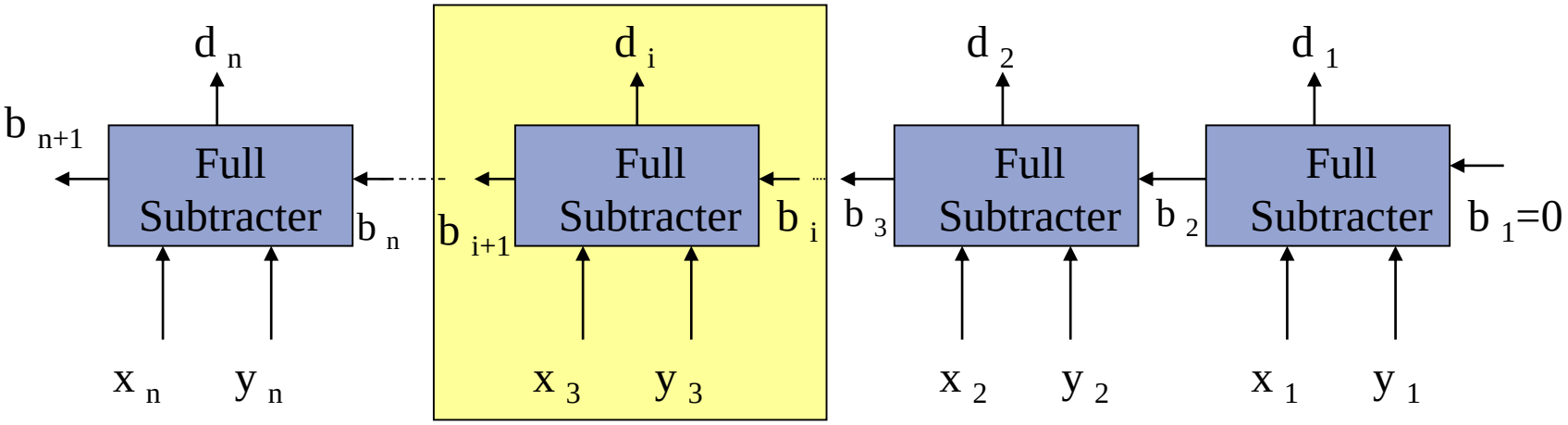
Subtracter of 2's complement system using full adders

Design of Binary Subtracter (II)

2) Design the full subtracter directly (X-Y)

=> use the truth table of inputs and outputs

=> consider **borrow** signals as input/output



Design of Binary Subtracter (III)

Truth Table for a Full Subtracter

$$d_i = x_i \oplus y_i \oplus b_i$$

$$b_{i+1} = b_i x_i' + x_i' y_i + b_i y_i$$

x_i	y_i	b_i	b_{i+1}	d_i
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$b_{i+1} = x'_i y'_i b_i + x'_i y_i b'_i + x'_i y_i b_i + x_i y_i b_i$$